

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 5/1/2025

Compounds: Loperamide, Pseudoephedrine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Loperaphedrine

Formula: C₄₈H₆₆N₂O₄

Weight: 719.07

Drug-likeness: 0.65

Activity:

Antidiarrheal and Nasal Decongestant

Target Analysis

Mu-Opioid Receptor

Binding Affinity: -9.3

Significant in modulating intestinal motility and secretion

Alpha-adrenergic Receptor

Binding Affinity: -8.1

Important for vasoconstriction and nasal congestion relief

Toxicity Profile

Risk Level: Moderate to High

Affected Systems: Heart, Central Nervous System, Liver

Mitigation Suggestions:

- Controlled administration
- Monitor cardiovascular function
- Avoid use with other CNS depressants

Pharmacokinetics

Absorption:

Moderate

Distribution:

Widely distributed with a high volume of distribution

Metabolism:

Extensively hepatic

Excretion:

Renal and fecal

Half-life: 10-12 hours

Detailed Analysis

In combining Loperamide, a peripherally acting opioid receptor agonist used for diarrheal conditions, with Pseudoephedrine, an alpha-adrenergic agonist used for nasal decongestion, the hybrid compound Loperaphedrine offers a dual-action therapeutic potential.

Loperaphedrine is designed to ameliorate gastrointestinal discomfort while simultaneously providing relief from nasal congestion, aiming to balance opioid and adrenergic activities. Its molecular structure is optimized for targeting both the mu-opioid and alpha-adrenergic receptors with predicted binding affinities of -9.3 and -8.1, respectively. While providing a valuable therapeutic duality, the compound poses moderate to high toxicity risks, particularly concerning cardiovascular and CNS effects, which necessitate controlled administration and careful monitoring during potential treatment courses. Further research and clinical trials are suggested to refine its efficacy and safety for broader application.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>